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APPLICATION NO.	FILING DATE	FIRST NAMED INVENTOR	ATTORNEY DOCKET NO.	CONFIRMATION NO.
17/570,917	01/07/2022	Adrian I. Bot	K-1104-US-NP	9051

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EXAMINER	
SAINE, LATIA D	

ART UNIT	PAPER NUMBER
1642	

NOTIFICATION DATE	DELIVERY MODE
04/05/2024	ELECTRONIC

Please find below and/or attached an Office communication concerning this application or proceeding.

The time period for reply, if any, is set in the attached communication.

Notice of the Office communication was sent electronically on above-indicated "Notification Date" to the following e-mail address(es):

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Office Action Summary	Application No. 17/570,917	Applicant(s) Bot et al.	
	Examiner LATIA D SAINE	Art Unit 1642	AIA (FITF) Status Yes

-- The MAILING DATE of this communication appears on the cover sheet with the correspondence address --

Period for Reply

A SHORTENED STATUTORY PERIOD FOR REPLY IS SET TO EXPIRE 3 MONTHS FROM THE MAILING DATE OF THIS COMMUNICATION.

- Extensions of time may be available under the provisions of 37 CFR 1.136(a). In no event, however, may a reply be timely filed after SIX (6) MONTHS from the mailing date of this communication.
- If NO period for reply is specified above, the maximum statutory period will apply and will expire SIX (6) MONTHS from the mailing date of this communication.
- Failure to reply within the set or extended period for reply will, by statute, cause the application to become ABANDONED (35 U.S.C. § 133). Any reply received by the Office later than three months after the mailing date of this communication, even if timely filed, may reduce any earned patent term adjustment. See 37 CFR 1.704(b).

Status

1)☑ Responsive to communication(s) filed on 2/22/2024.

☐ A declaration(s)/affidavit(s) under **37 CFR 1.130(b)** was/were filed on _____.

2a)☐ This action is **FINAL.** 2b) ☑ This action is non-final.

3)☐ An election was made by the applicant in response to a restriction requirement set forth during the interview on _____; the restriction requirement and election have been incorporated into this action.

4)☐ Since this application is in condition for allowance except for formal matters, prosecution as to the merits is closed in accordance with the practice under *Ex parte Quayle*, 1935 C.D. 11, 453 O.G. 213.

Disposition of Claims*

5) ☑ Claim(s) 1-17 is/are pending in the application.

5a) Of the above claim(s) 3-8 and 12-17 is/are withdrawn from consideration.

6) ☐ Claim(s) _____ is/are allowed.

7) ☑ Claim(s) 1-2 and 9-11 is/are rejected.

8) ☑ Claim(s) 1 is/are objected to.

9) ☐ Claim(s) _____ are subject to restriction and/or election requirement

* If any claims have been determined allowable, you may be eligible to benefit from the **Patent Prosecution Highway** program at a participating intellectual property office for the corresponding application. For more information, please see http://www.uspto.gov/patents/init_events/pph/index.jsp or send an inquiry to **PPHfeedback@uspto.gov.**

Application Papers

10)☑ The specification is objected to by the Examiner.

11)☑ The drawing(s) filed on 01/07/2022 is/are: a)☐ accepted or b)☑ objected to by the Examiner.

Applicant may not request that any objection to the drawing(s) be held in abeyance. See 37 CFR 1.85(a).

Replacement drawing sheet(s) including the correction is required if the drawing(s) is objected to. See 37 CFR 1.121(d).

Priority under 35 U.S.C. § 119

12)☐ Acknowledgment is made of a claim for foreign priority under 35 U.S.C. § 119(a)-(d) or (f).

Certified copies:

a)☐ All b)☐ Some** c)☐ None of the:

1.☐ Certified copies of the priority documents have been received.

2.☐ Certified copies of the priority documents have been received in Application No. _____.

3.☐ Copies of the certified copies of the priority documents have been received in this National Stage application from the International Bureau (PCT Rule 17.2(a)).

** See the attached detailed Office action for a list of the certified copies not received.

Attachment(s)

1) ☑ Notice of References Cited (PTO-892)

2) ☑ Information Disclosure Statement(s) (PTO/SB/08a and/or PTO/SB/08b) Paper No(s)/Mail Date 2/22/2024 and 2/22/2024.

3) ☐ Interview Summary (PTO-413) Paper No(s)/Mail Date _____.

4) ☑ Other: See Continuation Sheet.

Continuation of Attachment(s) 4) Other: BIB DATA SHEET filled on: 01/07/2022

DETAILED ACTION

Notice of Pre-AIA or AIA Status

The present application, filed on or after March 16, 2013, is being examined under the first inventor to file provisions of the AIA.

Election/Restrictions

Applicant's election without traverse of Group I and species of measuring the (B cells, T cells, and NK cells) lymphocytes to monocyte ratio, anti- inflammatory medications and small molecule inhibitors as two or a distinct set of drugs for the combination therapy and subspecies thereof, and lymphoma as the malignancy by tissue of origin (claims 1,2,9,10, and 11) in the reply filed on February 22, 2024 is acknowledged.

Claims 12-17 are withdrawn from further consideration pursuant to 37 CFR 1.142(b) as being drawn to a nonelected invention, there being no allowable generic or linking claim. Election was made **without** traverse in the reply filed on 2/22/2024.

Claims 3-8 are withdrawn from further consideration pursuant to 37 CFR 1.142(b) as being drawn to a nonelected species, there being no allowable generic or linking claim. Election was made **without** traverse in the reply filed on February 22,2024.

Claim Status

Claims 1-17 are pending.

Claims 1,2,9,10, and 11 are currently under examination for patentability under 37 CFR 1.104.

Priority

This application claims priority to U.S. Provisional Patent Application No. 63/135,711 filed January 10, 2021. Claims 1,2,9,10, and 11 have an effective filing date of **January 10, 2021**.

Information Disclosure Statement

The information disclosure statements (IDS) submitted on 2/22/2024 are in compliance with the provisions of 37 CFR 1.97. Accordingly, the information disclosure statements are being considered by the examiner.

Specification

1. The title of the invention is not descriptive. A new title is required that is clearly indicative of the invention to which the claims are directed. The title "T cell therapy" does not describe the invention.

2. The disclosure is objected to because it contains an embedded hyperlink and/or other form of browser-executable code at paragraph [0369]. Applicant is required to delete the embedded hyperlink and/or other form of browser-executable code; references to websites should be limited to the top-level domain name without any prefix such as http:// or other browser-executable code. See MPEP § 608.01.

Drawings

1. Color photographs and color drawings are not accepted in utility applications unless a petition filed under 37 CFR 1.84(a)(2) is granted. Any such petition must be accompanied by the appropriate fee set forth in 37 CFR 1.17(h), one set of color drawings or color photographs, as appropriate, if submitted via the USPTO patent electronic filing system or three sets of color drawings or color photographs, as appropriate, if not submitted via the via USPTO patent electronic filing system, and, unless already

present, an amendment to include the following language as the first paragraph of the brief description of the drawings section of the specification:

The patent or application file contains at least one drawing executed in color. Copies of this patent or patent application publication with color drawing(s) will be provided by the Office upon request and payment of the necessary fee.

Color photographs will be accepted if the conditions for accepting color drawings and black and white photographs have been satisfied. See 37 CFR 1.84(b)(2).

Finally, figures 1A-1B are not clear and legible. Applicant is therefore requested to provide clear and legible copies of such figures.

Claim Objections

3. Claim 1 is objected to because of the following informalities: The term "Th" contains an abbreviation and/or acronym that should be spelled out upon first occurrence. Appropriate correction is required.

Claim Rejections - 35 USC § 112(b)

4. The following is a quotation of 35 U.S.C. 112(b):

> (b) CONCLUSION. —The specification shall conclude with one or more claims particularly pointing out and distinctly claiming the subject matter which the inventor or a joint inventor regards as the invention.

The following is a quotation of 35 U.S.C. 112 (pre-AIA), second paragraph:

> The specification shall conclude with one or more claims particularly pointing out and distinctly claiming the subject matter which the applicant regards as his invention.

5. Claim 1,2,9,10, and 11 are rejected under 35 U.S.C. 112(b) or 35 U.S.C. 112 (pre-AIA), second paragraph, as being indefinite for failing to particularly point out and distinctly claim the subject matter

which the inventor or a joint inventor (or for applications subject to pre-AIA 35 U.S.C. 112, the applicant), regards as the invention.

6. The term “based on” in claim 1, 2, 9, and 10 is a relative term which renders the claim indefinite. The term “based on” is not defined by the claim, the specification does not provide a standard for ascertaining the requisite degree, and one of ordinary skill in the art would not be reasonably apprised of the scope of the invention. Notably, the determining step recites a single measurement that could form only a “part” of the decision-making process for determining administration, without identifying the other factors under consideration. Further, it is unclear whether the “determining” step is choosing whether to administer a composition, or choosing the type of treatment which gives the claims two interpretations, rendering the claims indefinite.

7. Regarding claim 1,2,9, and 10 the phrase “at least in part” renders the claim indefinite because it is unclear what decision part is required from the level of CD27+CD28+, and what other elements could be part of the decision. The entire set of factors used in the decision-making is not mentioned in the claim. If there is a minimum part for the level of CD27+CD28+ that is required to practice the method, it is not defined in the specification or the claims. This makes the claims unclear.

8. Regarding claims 2 and 10, the phrase “preferably” renders the claim indefinite because it is unclear whether the limitation(s) that are before the phrase are part of the claimed invention. See MPEP 2173.05(d).

9. A broad range or limitation together with a narrow range or limitation that falls within the broad range or limitation (in the same claim) may be considered indefinite if the resulting claim does not clearly set forth the metes and bounds of the patent protection desired. See MPEP § 2173.05(c). In the present instance, claim 11 recites the broad recitation of a genus of therapies including immunotherapies, chemotherapies, small molecule inhibitors, and the claim also recites antibodies targeted towards enhancing anti-tumor immunity, cyclophosphamide, fluoruracil, gemcitabine,

doxorubicin, and taxanes which are the narrower statements of the ranges/limitations. The claim(s) are considered indefinite because there is a question or doubt as to whether the feature introduced by such narrower language is (a) merely exemplary of the remainder of the claim, and therefore not required, or (b) a required feature of the claims.

Claim Rejections - 35 USC § 112(a)

10. The following is a quotation of the first paragraph of 35 U.S.C. 112(a):

> (a) IN GENERAL. —The specification shall contain a written description of the invention, and of the manner and process of making and using it, in such full, clear, concise, and exact terms as to enable any person skilled in the art to which it pertains, or with which it is most nearly connected, to make and use the same, and shall set forth the best mode contemplated by the inventor or joint inventor of carrying out the invention.

The following is a quotation of the first paragraph of pre-AIA 35 U.S.C. 112:

> The specification shall contain a written description of the invention, and of the manner and process of making and using it, in such full, clear, concise, and exact terms as to enable any person skilled in the art to which it pertains, or with which it is most nearly connected, to make and use the same, and shall set forth the best mode contemplated by the inventor of carrying out his invention.

Enablement

Claims 1,2,9,10, and 11 are rejected under 35 U.S.C. 112(a) or 35 U.S.C. 112 (pre-AIA), first paragraph, because the specification, while being enabling for treating patients with hematological cancers such as leukemia, lymphoma, and multiple myeloma with an effective dose of T cells comprising a chimeric antigen receptor (CAR), does not reasonably provide enablement for preventing any type of cancer types or treating any cancer except leukemia, lymphoma or multiple myeloma with all types of chimeric antigen receptors (CARs). The specification does not enable any person skilled in the art to which it pertains, or with which it is most nearly connected, to make and use the invention commensurate in scope with these claims.

It is noted that MPEP 2164.03 teaches that "the amount of guidance or direction needed to enable the invention is inversely related to the amount of knowledge in the state of the art as well as the predictability of the art. In re Fisher, 427 F.2d 833, 839, 166 USPQ 18, 24 (CCPA 1970). The amount of guidance or direction refers to that information in the application, as originally filed, that teaches exactly how to make or use the invention. The more that is known in the prior art about the nature of the invention, how to make, and how to use the invention, and the more predictable the art is, the less information needs to be explicitly stated in the specification. In contrast, if little is known in the prior art about the nature of the invention and the art is unpredictable, the specification would need more detail as how to make and use the invention in order to be enabling.".

As a general rule, enablement must be commensurate with the scope of claim language. MPEP 2164.08 states, "The Federal Circuit has repeatedly held that "the specification must teach those skilled in the art how to make and use the full scope of the claimed invention without undue experimentation'." In re Wright, 999 F.2d 1557, 1561, 27 USPQ2d 1510, 1513 (Fed. Cir. 1993)" (emphasis added). The "make and use the full scope of the invention without undue experimentation" language was repeated in 2005 in Warner-Lambert Co. v. Teva Pharmaceuticals USA Inc., 75 USPQ2d 1865, and Scripps Research Institute v. Nemerson, 78 USPQ2d 1019 asserts: "A lack of enablement for the full scope of a claim, however, is a legitimate rejection." The principle was explicitly affirmed most recently in Auto. Tech. Int'l, Inc. v. BMW of N. Am., Inc., 501 F.3d 1274, 84 USPQ2d 1108 (Fed. Cir. 2007), Monsanto Co. v. Syngenta Seeds, Inc., 503 F.3d 1352, 84 U.S.P.Q.2d 1705 (Fed. Cir. 2007), and Sitrick v. Dreamworks, LLC, 516 F.3d 993, 85 USPQ2d 1826 (Fed. Cir. 2008). See also In re Cortright, 49 USPQ2d 1464, 1466 and Bristol-Myers Squibb Co. v. Rhone-Poulenc Rorer Inc., 49 USPQ2d 1370.

Enablement is considered in view of the Wands factors (MPEP 2164.01 (A)). The factors considered when determining if the disclosure satisfies the enablement requirement and whether any

necessary experimentation is undue include, but are not limited to (In re Wands, 858 F.2d 731, 737, 8 USPQ2d 1400, 1404 (Fed. Cir. 1988)):

1) nature of the invention;

2) the breadth of the claims;

3) the state of the prior art;

4) the level of one of ordinary skill;

5) the level of predictability in the art;

6) the amount of direction or guidance provided by the inventor;

7) the existence of working examples; and

8) the quantity of experimentation needed to make or use the invention based on the content of the disclosure.

When the above factors are weighed, it is the examiner's position that one skilled in the art could not practice the invention without undue experimentation. Some experimentation is not fatal; the issue is whether the amount of experimentation is "undue"; see In re Vaeck, 20 USPQ2d 1438, 1444.

(1) The nature of the invention and (5) The breadth of the claims:

The claims are directed to a method for treating a malignancy in a patient comprising: measuring a level of CD27+CD28+ naive Th cells in an apheresis product from said patient; determining whether said patient should be administered an effective dose of T cells comprising a chimeric receptor, or an effective dose of T cells comprising a chimeric receptor and a combination therapy at least in part from said level of CD27+CD28+naive Th cells in said apheresis product; and administering said effective dose of T cells comprising a chimeric receptor, or said effective dose of T cells and said combination therapy based on said determining step, wherein said patient is administered said effective dose of T cells comprising a chimeric receptor if the level of CD27+CD28+ naive Th cells is over a cut-off percentage value measured as a percentage of total leukocytes, and wherein said patient is

administered said effective dose of T cells comprising a chimeric receptor and said combination therapy if the level of CD27+CD28+naive Th cells is below said cut-off percentage value.

The claims are overly broad and inclusive of all types of CAR molecules, which can target any of type of antigen molecule. Further, the claims are broad and inclusive of all types of cancer in humans generally. The breadth of the claim exacerbates the complex nature of the subject matter to which the present claims are directed. The claims are extremely broad due to the vast number of possible cancer types and tumor cell growth mechanisms represented by the phrase "malignancy". Cancer is not a single disease, or cluster of closely related disorders. There are hundreds of cancers, which have in common only some loss of controlled cell growth. Cancers are highly heterogeneous at both the molecular and clinical level, something seen especially in, for example, the cancers of the breast, brain and salivary glands. They can occur in pretty much every part of the body.

(2) The state of the prior art and (4) The predictability or unpredictability of the art:

While the state of the art is relatively high with regard to the treatment of specific cancer types, the state of the art with regard to treating cancer broadly is underdeveloped. In particular, there is no known anticancer agent that is effective against all cancer cell types. The cancer treatment art involves a very high level of unpredictability. While the state of the art is relatively high with regard to the treatment of specific cancers with specific agents, it has long been underdeveloped with regard to the treatment of cancers broadly. The lack of significant guidance from the present specification or prior art with regard to the actual treatment of all cancer cells in a mammal, including a human subject, with the claimed active ingredient makes practicing the claimed invention unpredictable.

Cancer cells arising from different tissues differ in etiology and response to treatment. Heppner et al. (Cancer Metastasis Review 2:5-23; 1983) discuss the heterogeneity of tumors from different tissues, as well as the same tissue. A key point made by Heppner et al. is that tumor heterogeneity contributes greatly to the sensitivity of tumors to drugs. Heppner et al. teach that as a tumor progresses

to a metastatic phenotype, the susceptibility to a particular treatment can differ, and as such, makes predicting the responsiveness to treatment difficult.

With regard to cancer treatment, Bally et al. (US 5,595,756) stated, "Despite enormous investments of financial and human resources, no cure exists for a variety of diseases. For example, cancer remains one of the major causes of death. A number of bioactive agents have been found, to varying degrees, to be effective against tumor cells. However, the clinical use of such antitumor agents has been highly compromised because of treatment-limiting toxicities" (col. 1, lines 17-24).

Sporn et al ("Chemoprevention of Cancer," Carcinogenesis, Vol. 21 (2000), 525-530), teaches the magnitude of mortality of cancers and those mortalities are in fact still rising and that new approaches to a variety of different cancer are critically needed. Sporn et al also teaches that "given the genotype and phenotype heterogeneity of advanced malignant lesions as they occur in individual patients, one wonders just exactly what are the specific molecular and cellular targets for the putative cure."

Furthermore, the art indicates the difficulties in going from *in vitro* to *in vivo* for drug development for treatment of cancers. Auerbach et al (Cancer and Metastasis Reviews, 2000, 19: 167-172) indicates that one of the major problems in angiogenesis research has been the difficulty of finding suitable methods for assessing the angiogenic response. For example, the 96 well rapid screening assay for cytokinesis was developed in order to permit screening of hybridoma supernatants…In vitro tests in general have been limited by the availability of suitable sources for endothelial cells, while in vivo assays have proven difficult to quantitate, limited in feasibility, and the test sites are not typical of the in vivo reality (see p. 167, left column, 1st paragraph). Gura T (Science, 1997, 278(5340): 1041-1042, encloses 1-5) indicates that "the fundamental problem in drug discovery for cancer is that the model systems are not predictive at all" (see p. 1, 2nd paragraph). Furthermore, Gura T indicates that the results of xenograft screening turned out to be not much better than those obtained with the original models, mainly because the xenograft rumors don't behave like naturally occurring tumors in humans—they

don't spread to other tissues, for example (see p. 2, 4th paragraph). Further, when patient's tumor cells in Petri dishes or culture flasks and monitor the cells' responses to various anticancer treatments, they don't work because the cells simply fail to divide in culture, and the results cannot tell a researcher how anticancer drugs will act in the body (see p. 3, 7th paragraph).

Furthermore, Jain RK (Scientific American, July 1994,58-65) indicates that the existing pharmacopoeia has not markedly reduced the number of deaths caused by the most common solid tumors in adults, among them cancers of the lung, breast, colon, rectum, prostate and brain (see p. 58, left most column, 1st paragraph). Jain RK indicates that to eradicate tumors, the therapeutic agents must then disperse throughout the growths in concentrations high enough to eliminate every deadly cells…solid cancers frequently impose formidable barriers to such dispersion (see p. 58, bottom of the left most column continuing onto the top of the middle column). Jain RK indicates that there are 3 critical tasks that drugs must do to attack malignant cells in a tumor: 1) it has to make its way into a microscopic blood vessel lying near malignant cells in the tumor, 2) exit from the vessel into the surrounding matrix, and 3) migrate through the matrix to the cells. Unfortunately, tumors often develop in ways that hinder each of these steps (see p. 58, bottom of right most column). Thus, the art recognizes that going from in vitro studies to in vivo studies for cancer drug developments are difficult to achieve.

CAR T cell therapy is also unpredictable, and recent studies demonstrate that many aspects of CAR signaling are unique, distinct from endogenous TCR signaling, and potentially even distinct among various CAR constructs (Science Advances, 2020 May, abstract). Thus, rigorous and comprehensive proteomic investigations are required for rational engineering of improved CARs (Lindner et al, Science Advances, 2020 May, abstract). CAR T cell therapy has transformed many aspects of clinical and translational oncology, and the successes achieved to date have led to a rapid expansion of clinical and basic science research efforts in this area (Lindner et al, Science Advances, 2020 May, page 6, par 2).

However, this explosion has, in many ways, outpaced fundamental investigations into how these receptors work, which threatens to limit our ability to improve upon current designs in an optimal fashion (Lindner et al, Science Advances, 2020 May, page 6, par 2). Characteristics such as CAR avidity, scFv affinity, antigen-binding domain structure and size, hinge/spacer region length and design, and transmembrane domain choice all affect the kinetics and dynamics of signaling pathway activation, just as choice of cytoplasmic signaling moieties affects the specific pathways that are proximally activated (Science Advances, 2020 May, page 6, par 2).

CAR-T cell therapies have been approved by the FDA to treat some kinds of leukemias and lymphomas, as well as multiple myeloma (see page 13, American Cancer Society "CAR T-cell therapy and its side effects", downloaded from https://www.cancer.org/cancer/managing-cancer/treatment-types/immunotherapy/car-t-cell1.html on 3/20/24; last revised December 20, 2023). At least six different therapies have been approved for use by the US Food and Drug Administration (FDA) in these cancers (see p. 13, American Cancer Society, entire reference). According to Guzman et al (Current Oncology Reports (2023) 25:479–489), CAR-T therapy has shown success in treating hematological malignancies (see e.g., page 480, left column, second paragraph).

Despite revolutionary progress in blood cancers, equivalent success has yet to be duplicated in solid tumor malignancies (see e.g., page 480, left column, second paragraph). Significant developments such as allogeneic or "of-the-shelf" CAR-T cells, which involve T cells from a donor's circulating blood, have attempted to improve solid tumor treatment success (see e.g., page 480, left column, second paragraph). However, solid tumors present unique challenges, such as a hostile tumor microenvironment (TME) and heterogeneous antigen expression that hinder similar advancements (see e.g., page 480, left column, second paragraph). A central difference between solid tumors and hematological cancers is the intricacy in detecting a perfect target antigen (see e.g., page 480, left column, third paragraph). While target heterogeneity exists in blood cancers, these disorders more

commonly tend to express individual markers (i.e., B-cell marker CD19) (see e.g., page 480, left column, third paragraph). Additionally, solid tumors more often have tumor-associated antigens (TAA), where this antigen is highly expressed on the tumor itself but also at low levels in normal tissues (see e.g., page 480, left column, third paragraph). Furthermore, solid tumors display TAA heterogeneity between tumor types (primary vs. metastatic) and patients with the same cancer (see e.g., page 480, left column, third paragraph). Consequently, a significant roadblock is designing a proper CAR-T cell with the capability of finding an ideal target antigen in solid tumors (see e.g., page 481, right column, second paragraph to page 482, left column, first paragraph). Once a CAR-T cell invades the endothelial barrier, it must continue to survive and persist in the inhospitable tumor microenvironment (TME) (see e.g., page 481, right column, second paragraph to page 482, left column, first paragraph). The TME presents a myriad of obstacles to CAR-T therapy, including the tumor stroma composed of suppressive cell types such as tumor-associated macrophages (TAMs) and myeloid-derived suppressor cells (MDSCs) and hypoxic conditions that hamper its execution (see e.g., page 481, right column, second paragraph to page 482, left column, first paragraph). Further, on-target off-tumor toxicity can be detrimental when CAR-T cells are directed to attack a tumor antigen that is also present in normal tissues (see e.g., page 482, left column, fourth paragraph). These data suggest that treatment of solid tumors or prevention of cancer with CAR-T cell therapy would be highly unpredictable and would require undue experimentation to identify working embodiments.

Hait (Nature Reviews/Drug Discovery, 2010, 9, pages 253-254) states that "The past three decades have seen spectacular advances in our understanding of the molecular and cellular biology of cancer. However, with a few notable exceptions, such as the treatment of chronic myeloid leukemia with imatinib, these advances have so far not been translated into major increases in long-term survival for many cancers. Furthermore, data suggest that the overall success rate for oncology products in clinical development is -10%, and the cost of bringing a new drug to market is over US$1 billion." (See

page 253, left column, the 1st paragraph). Hait further teaches "The anticancer drug discovery process often begins with a promising target; however, there are several reasons why the eventual outcome for a particular cancer target may be disappointing. For example, the role of the target in the pathogenesis of specific human malignancies may be incompletely understood, leading to disappointing results", "First, many targets lie within signal transduction pathways that are altered in cancer, but, owing to the complex nature of these pathways, upstream or downstream components may make modulating the target of little or no value"; "Second, target overexpression is often overrated. There are some instances in which overexpression predicts response to treatment"; and "Another confounding factor is that cancer is more than a disease of cancer cells, as alterations in somatic or germline genomes, or both, create susceptibilities to transformational changes in cells and in the microenvironment that ultimately cooperate to form a malignant tissue. Therefore, effective treatments may require understanding and disrupting the dependencies among the multiple cellular components of malignant tissues. The putative role of cancer stem cells in limiting the efficacy of cancer therapeutics is also an area of intense interest. Single nucleotide polymorphisms in genes responsible for drug metabolism can further complicate the picture by affecting drug pharmacokinetics; for example, as with the topoisomerase inhibitor irinotecan.", for example, page 253, Section "Understanding the target in context".

Beans (PNAS 2018; 115(50): 12539-12543) teaches that across cancer types, 90% of cancer deaths are caused not by the primary tumor but by metastasis. Beans teaches that although some drugs may shrink metastases along with primary tumors, no existing drugs treat or prevent metastasis directly (See page 12540). Beans states "Without a targeted approach, metastatic tumors often reemerge. "We shrink them, we send them back to their residual state, and they reenact those survival functions and retention of regenerative powers that made them metastasis-initiating cells in the first place" (See page 12540). Beans teaches that one of the major scientific challenges of studying metastatic disease is that different forms of cancer seem to metastasize through different mechanisms and the same form of

cancer may metastasize differently in different subsets of patients (See page 12542). Of note, Beans states “It’s unlikely that one researcher is going to find one pathway that proves to be the key to metastasis” (See page 12542). Beans also teaches that translating many findings into therapies also presents unique hurdles in that it is difficult to measure the effectiveness of the therapy. Secondary tumors are often minuscule, and therefore, measuring success by tumor shrinkage may not work. Measuring the incidence of metastasis after treatment is also more difficult (See page 12542).

Given Bally et al. teaching of treatment-limiting toxicities in clinical use; Sporn's teaching that the cancer progression is heterogeneous as it progresses, both in genotype and phenotype; Auerbach et al. teaching that one of the major problems in angiogenesis research has been the difficulty of finding suitable methods for assessing the angiogenic response; Gura's teaching that the models are unpredictable; Jain's teaching that the existing pharmacopoeia has not markedly reduced the number of deaths caused by the most common solid tumors in adults, among them cancers of the lung, breast, colon, rectum, prostate and brain; both Hait and Gravanis et al teaching various challenges facing cancer drug development, such as an understanding of cancer biology is far from complete, drug effects in preclinical cancer models often do not predict clinical results and many others; and Beans teachings that the field is highly underdeveloped with regards to preventing and treating cancer metastasis; the cited references demonstrate that the treatment of cancer is highly unpredictable, if even possible for many cancers.

6)The amount of direction or guidance provided by the inventor; 7) the existence of working examples:

The instant specification provides a clinical study wherein patients with relapsed/refractory non-Hodgkin’s lymphoma has been treated with axicabtagene ciloleucel was conducted. Axicabtagene ciloleucel is a CD19-directed genetically modified autologous T cell immunotherapy, comprising the patient's own T cells harvested and genetically modified ex vivo by retroviral transduction to express a

chimeric antigen receptor (CAR) comprising an anti-CD19 single chain variable fragment (scFv) linked to CD28 and CD3-zeta co-stimulatory domains (0366). However, the instant specification states that the cancer can be a leukemia, lymphoma or solid tumor or other type of cancer, meaning that virtually any type of cancer is encompassed by the instant claims (see e.g., specification paragraph [0012]). Given the evidence above and the heterogeneity of tumors, which have unique etiology and pathology, and unpredictable responsiveness to anti-cancer therapeutics.

Further, the specification defines the term "treating" to encompass preventing onset of cancer (see e.g., paragraph [0200]), which means that the claims read on cancer prevention. With respect to cancer prevention, there is no working example or other evidence that would provide the skilled artisan with any predictable guidance to make and use the claimed invention to prevent cancer without undue experimentation.

In conclusion, the claimed invention does not provide enablement for the prevention of cancer or treatment of non-hematological cancers with all types of CAR. Thus, for the reasons outlined above, the specification is not considered to be enabling for one skilled in the art to make and use the claimed invention as the amount of experimentation required is undue, due to the broad scope of the claims, the lack of guidance and working examples provided in the specification. Therefore, the specification is not representative of the instant claims and the specification is not fully enabled for the instant claims. In view of the above, one of skill in the art would be forced into undue experimentation to practice the claimed invention.

Written Description

11. Claims 1,2,9,10, and 11 are rejected under 35 U.S.C. 112(a) or 35 U.S.C. 112 (pre-AIA), first paragraph, as failing to comply with the written description requirement. The claim(s) contains subject matter which was not described in the specification in such a way as to reasonably convey to one skilled

in the relevant art that the inventor or a joint inventor, or for applications subject to pre-AIA 35 U.S.C. 112, the inventor(s), at the time the application was filed, had possession of the claimed invention.

The MPEP states that the purpose of the written description requirement is to ensure that the inventor had possession, as of the filing date of the application, of the specific subject matter later claimed. The MPEP lists factors that can be used to determine if sufficient evidence of possession has been furnished in the disclosure of the application. These include "level of skill and knowledge in the art, partial structure, physical and/or chemical properties, functional characteristics alone or coupled with a known or disclosed correlation between structure and function, and the method of making the claimed invention."

The written description requirement for a claimed genus may be satisfied through sufficient description of a representative number of species by actual reduction to practice, disclosure of drawings, or by disclosure of relevant identifying characteristics, for example, structure or other physical and/or chemical properties, by functional characteristics coupled with a known or disclosed correlation between function and structure, or by a combination of such identifying characteristics, sufficient to show the Applicants were in possession of the claimed genus.

The instant claims are drawn to a method for treating a malignancy in a patient comprising: measuring a level of CD27+CD28+ naive Th cells in an apheresis product from said patient; determining whether said patient should be administered an effective dose of T cells comprising a chimeric receptor, or an effective dose of T cells comprising a chimeric receptor and a combination therapy at least in part from said level of CD27+CD28+naive Th cells in said apheresis product; and administering said effective dose of T cells comprising a chimeric receptor, or said effective dose of T cells and said combination therapy based on said determining step, wherein said patient is administered said effective dose of T cells comprising a chimeric receptor if the level of CD27+CD28+ naive Th cells is over a cut-off percentage value measured as a percentage of total leukocytes, and wherein said patient is

administered said effective dose of T cells comprising a chimeric receptor and said combination therapy if the level of CD27+CD28+naive Th cells is below said cut-off percentage value. The method recites various determination factors of when and how to use the method in the patient.

The specification provides an example of administration of patients with relapsed/refractory NHL have been treated with axicabtagene ciloleucel was conducted (see instant specification paragraph [0366]). Axicabtagene ciloleucel is a CD 19-directed genetically modified autologous T cell immunotherapy, comprising the patient's own T cells harvested and genetically modified ex vivo by retroviral transduction to express a chimeric antigen receptor (CAR) comprising an anti-CD19 single chain variable fragment (scFv) linked to CD28 and CD3-zeta co-stimulatory domains (see e.g. instant specification paragraph [0366]). This was the only therapy that was tested. The instant specification also states that the population of the patients may have had diffuse large B-cell lymphoma, primary mediastinal B-cell lymphoma, or transformed follicular lymphoma with refractory disease despite undergoing recommended prior therapy [0367]. Patients received a target dose of 2x 106anti-CD19CAR T cells per kilogram of body weight after receiving a conditioning regimen of low-dose cyclophosphamide and fludarabine [0367]. However, the claims are far broader than the treatment methods and the CAR therapies supported by the examples.

There are at least three issues with regard to written description in the claims. First, the claims recite a CAR molecule that is not adequately described. The recited CAR is required to have specific functions, such as preventing and treating all types of malignancy, including solid tumors. These CAR molecules have not been identified, and one of skill in the art cannot immediately envisage the encompassed genus of CAR molecules. Further, no structure has been provided that will correlate with the required function, and a single tested species is not sufficient to represent such a vast genus of molecules, which must be useful for treatment and prevention of cancer based on measurement of CD27+CD28+ and potentially other markers.

The specification teaches a single example of CAR, which is antiCD-19 CAR T cells (see paragraph [0367] of the instant specification). However, the claims are not limited to this CAR. No other CAR T cells are provided. The CAR recited in the claims have no correlation between their structure and function. Further, the specification does not set forth a representative number of species for the breadth of the genus. Therefore, the specification provides insufficient written description to support the genus encompassed by the claim.

Second, the claims require a determination of therapy without adequately describing the steps that are necessary to determine the therapy. The claims recite that the therapy is determined "at least in part" based on the level of CD27+CD28+ naïve Th cells in an apheresis product, without fully defining the size of the "part" the CD27+CD28+ markers would contribute, or the additional factors necessary to make the determination. There is a vast array of signals that moderate CAR signaling—checkpoint molecules, immunosuppressive cytokines, oxygen tension, and other microenvironmental cues are all important for CAR T cell functions. There is no guidance in the specification to guide or identify the other parts of the determination step for a practitioner of the claimed invention. The claim reciting "at least in part" leaves it to the practitioner to discover which parts are necessary for the determination. Further, the steps do not define a "cut off percentage" and allows for 0-50%, which is an incredibly large range of values from which the practitioner must identify the appropriate values to practice the invention.

Third the "combination therapy" is not defined, and the specification does not provide criteria for identifying which therapies could perform the treatment functions of the claims, as a result of measurement of the encompassed biomarkers. The genus could encompass any number of small molecules, proteins, peptides, nucleic acids, antibodies or other molecules. While claim 11 names some genera of compounds, these compounds number in the thousands and one of skill in the art would not be able to immediate envisage the agents within these genera that would meet the requirements for the combination therapy that can treat all types of cancers in combination with the CAR molecules as

claimed. Further, many of the genera are defined only by their functions (e.g., “immunotherapies” “SRC kinase inhibitors” “TGF-beta inhibitors” various antibodies, etc.), without providing correlating structure for those functions.

Vas-Cath Inc. v. Mahurkar, 19 USPQ2d 1111, makes clear that

"Applicant must convey with reasonable clarity to those skilled in the art that, as of the filing date sought, he or she was in possession of the invention. The invention is, for purposes of the 'written description' inquiry, whatever is now claimed." (See page 1117.) The specification does not "clearly allow persons of ordinary skill in the art to recognize that [he or she] invented what is claimed." (See Vas-Cath at page 1116.)

University of California v. Eli Lilly and Co., 43 USPQ2d 1398, 1404. 1405 held that:

To fulfill the written description requirement, a patent specification must describe an invention and does so in sufficient detail that one skilled in the art can clearly conclude that "the inventor invented the claimed invention." Lockwood v. American Airlines Inc. , 107 F.3d 1565, 1572, 41 USPQ2d 1961, 1966 (1997); In re Gosteli , 872 F.2d 1008, 1012, 10 USPQ2d 1614, 1618 (Fed. Cir. 1989) (" [T]he description must clearly allow persons of ordinary skill in the art to recognize that [the inventor] invented what is claimed."). Thus, an applicant complies with the written description requirement "by describing the invention, with all its claimed limitations, not that which makes it obvious," and by using "such descriptive means as words, structures, figures, diagrams, formulas, etc., that set forth the claimed invention." Lockwood, 107 F.3d at 1572, 41 USPQ2datl966.

There are many challenges in predicting the success of CAR T- cell therapy. Traditional cancer treatments use nonspecific drugs and monoclonal antibodies to target tumor cells (See Dagar et al. Journal of Translational Medicine (2023) , at page 1 , abstract). Chimeric antigen receptor (CAR)-T cell therapy, however, leverages the immune system's T-cells to recognize and attack tumor cells (See Dagar, Abstract). Bi-specific chimeric antigen receptors may contribute to mitigating tumor antigen

escape, but their efficacy could be limited in cases where certain tumor cells do not express the targeted antigens (See Dagar, Abstract). CAR-T technology faces challenges in solid tumors, including lack of reliable tumor-associated antigens, hypoxic cores, immunosuppressive tumor environments, enhanced reactive oxygen species, and decreased T-cell infiltration (See Dagar, Abstract). Despite its benefits, it is still used as a second line of therapy, mainly because of its severe side effects and patient unresponsiveness (see Lindner et al., Sci. Adv. 2020, abstract).

Although existing predictive metrics correlate to some extent with treatment outcomes, they fail to encapsulate the complexity of the immune system dynamics. T (see Levestek, page 1, Introduction). Numerous researchers worldwide have attempted to identify effective predictive biomarkers for early prediction of treatment outcomes and adverse effects in CAR T cell therapy, so far with limited success (see Levestek, page 1, Introduction, and page 2, right column, second paragraph). To realize the full potential of CAR T cell therapy, a comprehensive understanding of the numerous factors influencing its efficacy is needed (see Levestek, page 14, par 3). In order to improve therapeutic efficacy, development of better biomarker models for predicting immune system response to CAR T cell infusion, cytotoxic efficacy of the infusion product, side effect susceptibility of each patient, therapeutic outcomes, and long-term remission is needed (see e.g., Levestek, page 2, right column, first paragraph). The multifactorial nature of therapeutic outcomes highlights the complexity and variability of any therapeutic success for CAR T cell therapy (see e.g., page 14, left column, second paragraph). Given the unpredictable nature of predicting success for CAR T-cell therapy, and for treatment of cancer in general, one of skill in the art would conclude that more evidence is necessary to show possession of the claimed genus of methods.

In particular, the requirement that one of skill in the art identify the “parts” of the determination step, the CAR T-cell that can treat a particular type of cancer, and a combination therapy that works in the claimed method, seems to be direction for a practitioner to actually determine what

the claimed method is, without providing guidance about how this would be accomplished. This seems to be expression of merely a wish or plan for the invention, without actually possessing the invention itself. The claimed method does little more than name a genus of methods with inadequately described steps that require the practitioner to perform a large amount of experimentation and optimization to practice any of the genus of methods that are encompassed. The specification does not provide sufficient guidance that would lead one of skill in the art to reasonably be able to practice the claimed method without need for further written description for the encompassed methods.

The claimed invention as a whole may not be adequately described where an invention is described solely in terms of a method of its making coupled with its function and there is no described or art-recognized correlation or relationship between the structure of the invention and its function (see MPEP 2163). A patent specification must set forth enough detail to allow a person of ordinary skill in the art to understand what is claimed and to recognize that the inventor invented what is claimed. In the case of DNA, an adequate written description requires a precise definition, such as by structure, formula, chemical name, or physical properties, not a mere wish or plan for obtaining the claimed chemical invention (see Lilly, 119 F.3d at 1566 (quoting Fiers, 984 F.2d 15 1171). Because the specification does not describe the in part of the combination therapy nor any core structures for potentially additional CAR therapies which would have the recited functions, one of skill in the art would reasonably conclude that applicant was not in possession of the claimed genus of CARs that treat and prevent all cancers.

Therefore, neither the art nor the specification provides a sufficient representative species or structures that correlate with the required functions for the claimed CAR therapy method to meet the written description requirements.

MPEP § 2163.02 states, “[a]n objective standard for determining compliance with the written description requirement is, 'does the description clearly allow person of ordinary skill in the art to

recognize that he or she invented what is claimed'". The courts have decided: the purpose of the "written description" requirement is broader than to merely explain how to "make and use"; the Applicant must convey with reasonable clarity to those skilled in the art, that as of the filing date sought, he or she was in possession of the invention. The invention is for purposes of the "written description" inquiry, whatever is now claimed. See Vas-Cath, Inc v. Mahurkar, 935 F.2d 1555, 1563-64, 19 USPQ2d 1111, 1117 (Federal Circuit, 1991).

Furthermore, the written description provision of 35 USC §112 is severable from its enablement provision; and adequate written description requires more than a mere statement that it is part of the invention and reference to a potential method for isolating it. Fiers v. Revel, 25 USPQ2d 1601, 1606 (CAFC 1993). And Amgen Inc. v. Chugai Pharmaceutical Co. Ltd., 18 USPQ2d 1016. Moreover, an adequate written description of the claimed invention must include sufficient description of at least a representative number of species by actual reduction to practice, reduction to drawings, or by disclosure of relevant, identifying characteristics sufficient to show that Applicant was in possession of the claimed genus. However, factual evidence of an actual reduction to practice has not been disclosed by Applicant in the specification; nor has Applicant shown the invention was "ready for patenting" by disclosure of drawings or structural chemical formulas that show that the invention was complete; nor has the Applicant described distinguishing identifying characteristics sufficient to show that Applicant were in possession of the claimed invention at the time the application was filed.

Therefore, for all these reasons the specification lacks adequate written description, and one of skill in the art cannot reasonably conclude that Applicant had possession of the claimed invention at the time the instant application was filed.

Claim Interpretation

The instant claims have been rejected under 35 USC 112(b) as being indefinite. The claims rely on an ambiguous step of determination "based on" measurement of CD27 and CD28, optionally along with any other possible biomarkers. The broadest reasonable interpretation will be applied for examination purposes. The determination step will be interpreted to read on selection of any type of therapy from CAR T-cell therapy and CAR T cell therapy and any combination, based on the level of CD27+CD28+ cells. The term "at least in part" will be interpreted as an open-ended phrase that encompasses an unlimited amount of other "parts" in the biomarker measurement step. Any measurement of CD27+CD28+ naïve T cells, even if other measurements are conducted, will read on the claims. The term "preferably" is interpreted as it will be ideal but any cut off percentage value in the claim will read on the claim. The term "based on" will be interpreted as the "determining" step that needs to be performed first before administering a composition, or choosing the type of treatment.

Claim Rejections - 35 USC § 103

In the event the determination of the status of the application as subject to AIA 35 U.S.C. 102 and 103 (or as subject to pre-AIA 35 U.S.C. 102 and 103) is incorrect, any correction of the statutory basis (i.e., changing from AIA to pre-AIA) for the rejection will not be considered a new ground of rejection if the prior art relied upon, and the rationale supporting the rejection, would be the same under either status.

The following is a quotation of 35 U.S.C. 103 which forms the basis for all obviousness rejections set forth in this Office action:

> A patent for a claimed invention may not be obtained, notwithstanding that the claimed invention is not identically disclosed as set forth in section 102, if the differences between the claimed invention and the prior art are such that the claimed invention as a whole would have been obvious before the effective filing date of the claimed invention to a person having ordinary skill in the art to which the

claimed invention pertains. Patentability shall not be negated by the manner in which the invention was made.

The factual inquiries for establishing a background for determining obviousness under 35 U.S.C. 103 are summarized as follows:

1. Determining the scope and contents of the prior art.

2. Ascertaining the differences between the prior art and the claims at issue.

3. Resolving the level of ordinary skill in the pertinent art.

4. Considering objective evidence present in the application indicating obviousness or nonobviousness.

This application currently names joint inventors. In considering patentability of the claims the examiner presumes that the subject matter of the various claims was commonly owned as of the effective filing date of the claimed invention(s) absent any evidence to the contrary. Applicant is advised of the obligation under 37 CFR 1.56 to point out the inventor and effective filing dates of each claim that was not commonly owned as of the effective filing date of the later invention in order for the examiner to consider the applicability of 35 U.S.C. 102(b)(2)(C) for any potential 35 U.S.C. 102(a)(2) prior art against the later invention.

Claim(s) 1,2, and 11 are rejected under 35 U.S.C. 103 as being unpatentable over PRUTEANU-MALINICI et al (WO2019213282 published 07 November 2019, cited and filed on an IDS 1449) in view of Worel et al (Blood (2019) 134 (Supplement_1); published November 13,2019, pgs. 1-7).

The instant claims are directed to a method for treating a malignancy in a patient comprising: measuring a level of CD27+CD28+ naive Th cells in an apheresis product from said patient; determining whether said patient should be administered an effective dose of T cells comprising a chimeric receptor, or an effective dose of T cells comprising a chimeric receptor and a combination therapy at least in part

from said level of CD27+CD28+naive Th cells in said apheresis product; and administering said effective dose of T cells comprising a chimeric receptor, or said effective dose of T cells and said combination therapy based on said determining step, wherein said patient is administered said effective dose of T cells comprising a chimeric receptor if the level of CD27+CD28+ naive Th cells is over a cut-off percentage value measured as a percentage of total leukocytes, and wherein said patient is administered said effective dose of T cells comprising a chimeric receptor and said combination therapy if the level of CD27+CD28+naive Th cells is below said cut-off percentage value. The combination therapies can include immunotherapies, which the instant specification states can include anti-PD-1 therapies (see e.g., paragraph [0017]).

PRUTEANU-MALINICI et al disclosure relates to the identification and use of biomarkers with clinical relevance cancer (page 3, paragraph 2). More specifically, PRUTEANU-MALINICI et al discloses methods, compositions, and kits for the identification, assessment and/or treatment of a subject having cancer (page 3, paragraph 4). PRUTEANU-MALINICI et al teaches a method of treating cancer comprising administering a CAR-expressing cell therapy to a subject having a cancer, comprising: measuring the level or activity of CD27 in immune effector cells, in an apheresis sample (page 4, paragraph 4, claim 7). PRUTEANU-MALINICI et al teaches administering an altered dosing of a CAR-expressing cell therapy and adjusting the CAR-expressing cell infusion dose to achieve clinical efficacy based on responder status (page 6, paragraph 4-5). Subjects can be responders, partial responders, or non-responders (see e.g., claims 1-7). PRUTEANU-MALINICI et al teaches in some embodiments of any of the methods and compositions for use disclosed herein, the level or activity of CD27+ immune effector cells are evaluated using a profile or signature indicative of the percentage of CD27+ immune effector cells in the sample (Page 30, paragraph 1). Non-responders or partial responders are defined as, while responders are identified as. Non-responders are administered a CAR T cell therapy and an additional agent, such as a checkpoint inhibitor (see e.g., claim 7). Checkpoint inhibitors include PD1 inhibitors (see e.g., page 218,

first paragraph). Responders are administered a CAR-expressing cell therapy (see e.g., claim 7). The responders are characterized as having greater percentage of CD27+ cells, such as 5% or more (see e.g., page 31, sixth and seventh paragraphs) compared to a reference value or cut off value (see e.g., page 82, fourth paragraph; claims 13-15). PRUTEANU-MALINICI teaches the apheresis product typically contains lymphocytes, including T cells, monocytes, granulocytes, B cells, other nucleated white blood cells, red blood cells, and platelets. In one aspect, the cells collected by apheresis may be washed to remove the plasma fraction and, optionally, to place the cells in an appropriate buffer or media for subsequent processing steps. In one embodiment, the cells are washed with phosphate buffered saline (PBS) (page 167, paragraph 3). The CAR that is administered can be a CD19 CAR (see e.g., page 156, first paragraph). The method can treat diffuse large B-cell lymphoma (see e.g., page 200, second paragraph).

With respect to claim 2, PRUTEANU-MALINICI et al teaches in some embodiments of any of the methods and compositions for use disclosed herein, a complete responder has, or is identified as having, a greater percentage e.g., 5%, 6%, 7%, 10%, 15%, 20%, 25%, 27%, 30%, 35%, or 40% or greater number of CD27+ cells (page 32, paragraph 2-5).

PRUTEANU-MALINICI et al does not teach the use of CD28+ measurement to determine responder status for selecting therapy.

Worel et al describes therapies involving adoptive transfer of chimeric antigen receptor-modified T-cells (CARTs) targeting CD19-expressing B-cells that have shown remarkable efficacy in patients with diffuse large B-cell lymphoma (DLBCL) (page 1, paragraph 1). This document discloses influence of CD27- CD28- T-Cells on the therapeutic outcome in adult patients with relapsed or refractory Diffuse Large B-Cell Lymphoma after CART-Infusion (see e.g., title). The reference further discloses peripheral blood and apheresis samples of 22 consecutive DLBCL patients scheduled for CART therapy were analyzed by flow cytometry to assess their CD27 and CD28 expression status on CD3 T-cells (see e.g., page 2, paragraph 1). The examples ruled out an apheresis-related bias within cell

populations when peripheral blood and apheresis samples were analyzed for each patient. Worel et al demonstrates in this small patient cohort that individuals with a lower percentage of more differentiated, senescent or exhausted T-cells are more likely to respond to CART therapy (see e.g., page 2, paragraph 3). A significantly lower frequency of CD27-/CD28- T cells (20.8±18.8 vs 42.4±13.7%; p=0.045 Mann Whitney U) were found in responders compared to non-responders (see e.g., Results section). Worel et al teach higher frequency of CD27+/CD28+ T-cells (57.5±22.7% vs 44.3±11.7%; Figure 1b) and a trend towards a normal (>1.0) CD4/CD8 ratio (Figure 1c) in responding patients. Worel et al underscores the importance of T-cells with normal replicative capacity in the apheresis material for consecutive CART production to achieve therapeutic success (see e.g., page 2, paragraph 3).

It would have been prima facie obvious to one of ordinary skill in the art at the time of the invention to modify the administration and selection based on CD27+ cells of Pruteanu-Malinici to include Worel's method of testing for both CD27+ and CD28+ cells to determine responder status for selecting therapies involving adoptive transfer of chimeric antigen receptor-modified T-cells (CARTs) targeting CD19-expressing B-cells (page 1, paragraph 1). The method of Worel shows that higher percentage of CD27+CD28+ cells will predict greater response to single therapy with CAR T-cells. The specific advantages of the process of Worel, including being able to identify individuals with a higher or lower percentage of normal replicative capacity to increase the chances of therapeutic success (page 2, conclusion) provides the motivation to combine the teachings of the references. One of skill in the art could modify the process of Pruteanu-Malinici to include the additional CD28 marker of Worel for the differential diagnosis and treatment of cancer with CAR-expressing cell therapy with a reasonable expectation of success.

Claim(s) 9 and 10 are rejected under 35 U.S.C. 103 as being unpatentable over PRUTEANU-MALINICI et al(cited and filed on an IDS 1449) in view of Worel et al (Blood (2019) 134

(Supplement_1); published November 13,2019, pgs. 1-7) in further view of Wang et al (Oncotarget, Vol. 8, (No.3), published December 27, 2016, pgs. 5414-5425).

The combination of PRUTEANU-MALINICI and Worel makes claims 1, 2, and 11 obvious for the reasons supra, all incorporated here.

Neither PRUTEANU-MALINICI nor Worel teach measuring a lymphocyte to monocyte ratio in a baseline hematology count of said patient; determining whether said patient should be administered an effective dose of T cells comprising a chimeric receptor, or an effective dose of T cells comprising a chimeric receptor and a combination therapy at least in part from said lymphocyte to monocyte ratio; and administering said effective dose of T cells comprising a chimeric receptor, or said effective dose of T cells and said combination therapy based on said determining step, wherein said patient is administered said effective dose of T cells comprising a chimeric receptor if the lymphocyte to monocyte ratio is above a cut-off value, and wherein said patient is administered said effective dose of T cells comprising a chimeric receptor and said combination therapy if the lymphocyte to monocyte ratio is below said cut-off value. Neither reference teaches the cut-off value of claim 10.

With respect to claims 9 and 10, Wang et al teaches lymphocyte to monocyte ratio is associated with prognosis of diffuse large B-cell lymphoma (DLBCL; see e.g., title). Wang et al teaches that diffuse large B- cell lymphoma is a common type of lymphoma, occupying thirty percent to forty of preliminary diagnosed non-Hodgkin's lymphomas, which can be cured with standard immunochemotherapy (see e.g., page 5414, paragraph 1). Wang et al teach several studies has showed that tumor-associated macrophages (TAM) and peripheral blood monocytes could inhibit host antitumor immunity and affect the prognosis of DLBCL (Diffuse large B-cell lymphoma) (see e.g., page 5415, paragraph 1). Wang et al teaches the measurement of lymphocyte to monocyte ratio (LMR) for assessing prognosis in DLBCL patients (see e.g., abstract). Decreased LMR leads to weak anti-tumor immunity and could be used as a

bad prognosis biomarker of DLBCL (see e.g., abstract). Wang et al cut-off values for lymphocyte to monocyte ratio (LMR) can be 2.71 (see e.g., page 5415, right column, paragraph 4) Wang et al also teach that potential of peripheral blood tests as surrogate biomarker of host immune microenvironment in DLBCL was also determined (see e.g., page 5415, left column, paragraph 2). Lastly, Wang teaches the pre- and post-treatment amounts of monocytes, and lymphocytes in peripheral blood, which were related to immune system and the prognosis of lymphomas see e.g., page 5419, left column, left column, p. 2).

It would have been obvious to one of ordinary skill in the art at the time of the invention to modify the administration and selection based on Pruteanu-Malinici and Worel's to include the measurement of lymphocyte to monocyte ratio because the measurement of this ratio can indicate antitumor immunity and prognosis (see e.g., abstract). The specific advantages of Wang are to use the value of the lymphocyte to monocyte ratio to increase antitumor immunity and prognosis in patients (see e.g., abstract). Wang also teach the pre- and post-treatment amounts of monocytes, neutrophils, and lymphocytes in peripheral blood, which were related to immune system and the prognosis of Lymphomas and solid tumors (see e.g., Wang, [pg. 5419, discussion, par. 1]). Further, Wang discusses the correlation between Lymphocyte to monocyte ratio and PD-1+ tumor infiltrating lymphocytes (TIL), which is a prognostic indicator (see e.g., page 5415, paragraph 3; abstract). Wang teaches that the number of PD-1+ TILs reflect tumor-mediated T-cell exhaustion (see e.g., page 5415, second paragraph), which was discussed as an important factor for selecting CAR T cell therapy regimens according to Worel. Applying the method of Wang to the method of Pruteanu-Malinici and Worel will provide greater indication of anti-tumor immunity, and offer greater evidence for selection of patients for anti-PD1 therapy as indicated by Pruteanu-Malinici. The advantages of adding an additional prognostic factor that can offer additional information regarding anti-tumor immunity, prognosis and tumor-mediated T cell exhaustion, which can lead to more accurate selection of therapy, provides the motivation to make the

aforementioned modification of the process of Worel to use the process with Pruteanu-Malinici and Wang with a reasonable expectation of success of treatment in cancer.

Additionally, KSR International Co. v. Teleflex Inc., 127 S. Ct. 1727, 1741 (2007), discloses that if a technique has been used to improve one method, and a person of ordinary skill would recognize that it would be used in similar methods in the same way, using the technique is obvious unless its application is beyond that person's skill. It would be obvious to apply a known technique to a known product to be used in a known method that is ready for improvement to yield predictable results. Thus, the combination of prior art references as combined provided a prima facie case of obviousness, absent convincing evidence to the contrary.

Conclusion

No claim is allowed.

Any inquiry concerning this communication or earlier communications from the examiner should be directed to LATIA DENISE SAINE whose telephone number is (703)756-1690. The examiner can normally be reached Monday-Friday 830am-500pm.

Examiner interviews are available via telephone, in-person, and video conferencing using a USPTO supplied web-based collaboration tool. To schedule an interview, applicant is encouraged to use the USPTO Automated Interview Request (AIR) at http://www.uspto.gov/interviewpractice.

If attempts to reach the examiner by telephone are unsuccessful, the examiner's supervisor, **571-270-3503** can be reached on **Samira Jean- Louis**. The fax phone number for the organization where this application or proceeding is assigned is 571-273-8300.

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